# Harry Nguyen

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## Education

**Georgia State University** Expected May 2027

Bachelor of Science in Computer Science · **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

## Work Experience

**Google** | Software Engineer Intern — Chrome Browser May 2025 – Aug 2025

- Profiled settings navigation as p99 bottleneck at 1,200ms; designed **lock-free concurrent trie search** eliminating mutex contention — **96% latency reduction**, zero production regressions in Chrome stable channel
- Designed **C++ IPC transport layer** with Protocol Buffers serving **3B+ active users** at **sub-50ms p99**, **10K+ req/sec** — selected Protobuf over custom serialization for schema evolution and zero-copy cross-process message passing
- Architected event-driven **TypeScript/React** observer system decoupling UI state propagation across **25K+ lines** of Chromium at **95% test coverage**; reviewed and adopted into production branch by senior Chrome engineers

**CoderPush** | Software Engineer Intern Sep 2023 – Dec 2023

- Diagnosed **DynamoDB hot-partition failure** in high-throughput data pipeline at **9,000+ req/sec**; redesigned partition key strategy and access patterns — **30% read throughput improvement** under sustained concurrency
- Designed idempotent **REST APIs** with **Redis** distributed caching (**85% cache hit rate**) and exponential backoff — enforced exactly-once semantics preventing duplicate transactions under network partitions

**Develop for Good** | Software Engineer Intern May 2024 – Aug 2024

- Designed stateless BaaS on **AWS** with JWT-based auth, auto-scaling supporting **500+ concurrent users**; diagnosed N+1 query bottleneck, redesigned **PostgreSQL** indexing achieving **sub-100ms** for 10,000+ records

## Personal Projects

**TiMoto AI – Concurrent Systems Platform** | *C++, Python, gRPC, Django, AWS, Terraform* timoto.ai

- Resolved **production deadlock under concurrent gRPC calls** as **sole engineer** — traced root cause via structured logging to circular resource acquisition; redesigned call sequencing with explicit lock ordering — **100% success rate** at **sub-50ms p99**
- Selected **vLLM** with **PagedAttention** for zero OOM failures under concurrent load — continuous batching and memory-aware scheduling manage KV cache under high concurrency without page faults
- Architected multi-AZ **ECS Fargate** with circuit breaker pattern and health-check-triggered auto-rollback — **99.9% uptime**, **44% cost reduction** to $40–60/month under production traffic

**Pulumi – Open Source Contributor** | *Go, TypeScript, IaC* 24.4K★ github.com/pulumi/pulumi

- Contributed **Go** CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning — under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer — studied linearizability guarantees and correctness proofs under concurrent operations and partial failures

## Technical Skills

**Languages**: C++, Python, Go, TypeScript, Java, JavaScript, Rust, SQL, Bash
**Distributed Systems & Concurrency**: gRPC, Protocol Buffers, Lock-free data structures, Exactly-once semantics, Circuit breakers, Raft/Paxos, Linearizability, Fault tolerance
**Algorithms & Performance**: Concurrent trie search, Hash-based partitioning, Mutex contention analysis, p99 profiling, Memory-aware scheduling
**Cloud & Infrastructure**: AWS (ECS Fargate, EC2, S3, RDS, DynamoDB), Kubernetes, Docker, Terraform, CI/CD, GitHub Actions
**ML & AI Infrastructure**: vLLM, PagedAttention, Continuous Batching, LLM-as-a-judge evaluation, PyTorch